

From Temperature Sensitivities to Melt-Pool Geometry Sensitivities

1 Liquidus points that define the geometry

Let p denote one process parameter,

$$p \in \{P, \sigma, v\}, \quad (1)$$

and let $T(\mathbf{x}, p)$ be the temperature field for that parameter value. The liquidus isotherm is the level set

$$T(\mathbf{x}, p) = T_{\text{liq}}. \quad (2)$$

The melt-pool width is defined by two geometric extrema of this isotherm:

- $y^-(p)$ is the leftmost liquidus point;
- $y^+(p)$ is the rightmost liquidus point.

Their locations depend on p because changing the process parameter changes the temperature field and therefore moves the liquidus boundary. Nevertheless, both moving points remain on the same liquidus isotherm:

$$T(\mathbf{x}^-(p), p) = T_{\text{liq}}, \quad (3)$$

$$T(\mathbf{x}^+(p), p) = T_{\text{liq}}. \quad (4)$$

The superscript $\pm$ identifies a spatial extremum; it is not a temperature maximum or minimum. Both points have temperature T_{liq} .

2 Motion of one liquidus edge

To expose the essential calculation, suppress the other spatial coordinates and write the right-edge condition as

$$T(y^+(P), P) = T_{\text{liq}}. \quad (5)$$

The material is fixed, so T_{liq} does not depend on P . Taking the total derivative of Equation (5) gives

$$\frac{d}{dP} T(y^+(P), P) = 0. \quad (6)$$

Applying the chain rule,

$$T_y|_{y^+} \frac{\partial y^+}{\partial P} + T_P|_{y^+} = 0, \quad (7)$$

where

$$T_y = \frac{\partial T}{\partial y}, \quad T_P = \frac{\partial T}{\partial P}. \quad (8)$$

Solving for the motion of the right edge gives

$$\boxed{\frac{\partial y^+}{\partial P} = - \left. \frac{T_P}{T_y} \right|_{y^+}}. \quad (9)$$

The same calculation at the left edge gives

$$\boxed{\frac{\partial y^-}{\partial P} = - \left. \frac{T_P}{T_y} \right|_{y^-}}. \quad (10)$$

The signs of T_y will normally differ at the two edges, but Equations (9) and (10) have the same form.

3 From edge motion to width sensitivity

The controller uses the melt-pool *half-width*

$$w = \frac{y^+ - y^-}{2}. \quad (11)$$

Differentiating Equation (11) with respect to P gives

$$\frac{\partial w}{\partial P} = \frac{1}{2} \left(\frac{\partial y^+}{\partial P} - \frac{\partial y^-}{\partial P} \right). \quad (12)$$

Substituting Equations (9) and (10),

$$\frac{\partial w}{\partial P} = \frac{1}{2} \left[- \left. \frac{T_P}{T_y} \right|_{y^+} + \left. \frac{T_P}{T_y} \right|_{y^-} \right] \quad (13)$$

$$= \boxed{\frac{1}{2} \left[\left. \frac{T_P}{T_y} \right|_{y^-} - \left. \frac{T_P}{T_y} \right|_{y^+} \right]}. \quad (14)$$

The factor $1/2$ appears because w is a half-width. If the reported quantity were the full width $W = y^+ - y^-$, this factor would be absent.

4 Physical interpretation of the ratio

At a fixed spatial point,

- T_P measures the temperature change produced by a change in power;
- T_y measures the temperature change produced by moving in the transverse direction.

Therefore,

$$- \frac{T_P}{T_y} \quad (15)$$

converts a parameter-induced temperature change into the spatial displacement required for the point to remain at $T = T_{\text{liq}}$. Sampling this ratio at the two liquidus edges and combining their motions produces the width sensitivity.

5 General process parameter and depth sensitivity

Replacing P by any $p \in \{P, \sigma, v\}$ gives

$$\boxed{\frac{\partial w}{\partial p} = \frac{1}{2} \left[\frac{T_p}{T_y} \Big|_{y^-} - \frac{T_p}{T_y} \Big|_{y^+} \right]}. \quad (16)$$

Let $z^-(p)$ be the deepest liquidus point and define the positive depth by

$$d = -z^-. \quad (17)$$

Implicit differentiation of the liquidus condition at z^- gives

$$\frac{\partial z^-}{\partial p} = - \frac{T_p}{T_z} \Big|_{z^-}. \quad (18)$$

Consequently,

$$\boxed{\frac{\partial d}{\partial p} = \frac{T_p}{T_z} \Big|_{z^-}}. \quad (19)$$

Equations (16) and (19), evaluated for $p = P, \sigma, v$, form the 2×3 geometry Jacobian used by the controller.

6 Assumptions

The formulas assume that the relevant liquidus extremum is locally smooth and unique, and that the required spatial gradient is nonzero: $T_y \neq 0$ at the width edges and $T_z \neq 0$ at the deepest point. In the full three-dimensional notation, the moving-point condition is

$$T(\mathbf{x}^*(p), p) = T_{\text{liq}}, \quad (20)$$

and its total derivative is

$$\nabla T \cdot \frac{d\mathbf{x}^*}{dp} + T_p = 0. \quad (21)$$

At a smooth support point, the tangency/extremum condition reduces this vector relation to the one-dimensional edge and depth expressions used above.